

Notes: Feldhütter & Schaefer (JFE, 2023)

Debt dynamics and credit risk

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1 Big picture

- **Question.** How do corporate debt-policy dynamics affect default risk and corporate bond pricing?
- **Core critique.** Most structural credit-risk models treat debt, or the default boundary, as constant or deterministic. That misses one of the two state variables that determine default: firm value and the default boundary.
- **Main empirical fact.** Corporate debt issuance has a sizable stochastic component that is not perfectly tied to asset-value shocks. In the short run, firms hit by negative equity shocks often take on more debt; in the long run, debt mean reverts toward target leverage.
- **Short-run vs. long-run pattern.** At short horizons, leverage volatility is higher than asset volatility because debt shocks add another source of risk. At long horizons, leverage mean reversion eventually lowers leverage volatility relative to asset volatility.
- **New model.** The paper proposes a stochastic-debt (SD) model: debt is mean reverting as in Collin-Dufresne and Goldstein (2001), but debt adjustment also has a stochastic shock that is negatively correlated with firm-value shocks.
- **Data.** The debt-dynamics evidence uses CRSP/Compustat industrial firms over 1988–2017. The bond-pricing exercise uses U.S. corporate bond spreads over 1988–2018, with bond data from Lehman, Merrill Lynch, TRACE, FISD, CRSP, Compustat, and Moody’s defaults.
- **Main calibration.** The models are calibrated to historical default rates, not to observed credit spreads. Thus, the spread tests are out-of-sample with respect to spreads.
- **Pricing result.** The SD model fits corporate spreads better in the cross-section and the time series, especially for short-maturity investment-grade bonds.
- **Mechanism.** Higher short-run leverage volatility forces the calibrated default boundary lower. This makes spreads less sensitive to observed leverage, raising spreads for safer firms and lowering spreads for riskier firms relative to standard diffusion models.
- **Additional evidence.** Adding stochastic debt to a stochastic-volatility/jump model improves the fit to Walmart CDS curves, reducing RMSEs substantially.
- **Economic takeaway.** Credit risk is not only about firm-value risk. Debt-policy risk is itself a priced component of credit risk because it changes the distribution of future leverage.

2 Data and measurement

2.1 Debt-level data and leverage

- The empirical debt-dynamics sample uses CRSP/Compustat for U.S. industrial firms, excluding utilities and financials.
- The sample period is 1988–2017 for the main debt-dynamics analysis, chosen to align with the corporate bond sample.
- The nominal amount of debt is debt in current liabilities plus long-term debt.
- The empirical proxy for firm value is market equity plus book debt.
- Leverage is measured as

$$L_{i,t} = \frac{D_{i,t}}{E_{i,t} + D_{i,t}},$$

where $D_{i,t}$ is book debt and $E_{i,t}$ is market equity.

- The paper restricts the debt-growth analysis to observations with positive debt.

Interpretation. This leverage measure is practical and observable for a large panel. It is not a perfect market-value leverage measure, so the paper later checks robustness using market values of debt.

2.2 Debt-growth measurement

For firm i , year t , and horizon T , the paper measures log debt growth as

$$R_{i,t}^T = \log \left(\frac{D_{i,t+T}}{D_{i,t}} \right).$$

The average T -year debt growth rate is

$$R^T = \frac{1}{N_T} \sum_i \sum_t R_{i,t}^T.$$

Interpretation. The log ratio is less sensitive to outliers than a simple ratio and matches the log-debt dynamics used in the structural models.

2.3 Conditioning on equity returns

- Within each initial leverage bin, the paper sorts firms by future three-year equity returns.
- Firms above the median return in their leverage bin are labeled high-equity-return firms; firms below the median are labeled low-equity-return firms.
- The authors then compare debt growth before, during, and after the three-year equity-return window.

Interpretation. Future equity returns proxy for asset-value shocks. The key question is whether debt moves with the shock, against the shock, or with a lag toward target leverage.

2.4 Corporate bond data

- The bond sample contains non-callable fixed-coupon bonds issued by industrial firms.
- The paper combines monthly quote data from Lehman Brothers and Merrill Lynch with TRACE transaction data.
- Bond characteristics come from FISD.
- The sample restricts to senior unsecured fixed-rate or zero-coupon bonds, excludes callable, convertible, puttable, perpetual, Yankee, foreign-currency, and sinking-fund bonds, and restricts maturities to under 20 years.
- Spreads are mainly measured relative to swap rates.

Interpretation. These filters are designed to make the observed yield spread closer to a clean credit-risk spread, rather than a mixture of optionality, currency risk, or unusual contractual features.

2.5 Firm variables used in pricing

For each bond-spread observation, the paper constructs:

- firm leverage;
- equity volatility from daily stock returns over the prior three years;
- asset volatility by unlevering equity volatility;
- payout ratio, measured as interest payments plus dividends plus repurchases divided by firm value.

Interpretation. The pricing models require firm-level leverage, volatility, and payout inputs. Asset volatility is central because it determines the volatility of the firm-value process.

2.6 Historical default rates

- The paper calibrates default boundaries to historical default rates for U.S. industrial firms.
- The default data come from Moody’s Default and Recovery Database.
- The authors use default rates over 1970–2017 and emphasize that U.S. industrial default rates differ from global Moody’s rates.
- C-rated firms are excluded when estimating default boundaries in the main specification because book debt can distort leverage for distressed firms.

Interpretation. Matching default rates before testing spreads separates default-probability calibration from the pricing test. It also avoids fitting model parameters directly to spreads.

3 Models and empirical specifications

3.1 Firm value and default

All four baseline diffusion models assume that firm value follows a geometric Brownian motion under the physical measure:

$$\frac{dV_t}{V_t} = (\mu - \delta)dt + \sigma dW_t,$$

where μ is the expected asset return, δ is the payout rate, and σ is asset volatility.

Default occurs when firm value hits a fraction d of face debt:

$$\tau = \inf\{t : V_t \leq dK_t\}.$$

Using $v_t = \log V_t$, $k_t = \log K_t$, and $l_t = k_t - v_t$, this becomes

$$\tau = \inf\{t : l_t \geq -\log(d)\}.$$

Interpretation. Default depends on the path of leverage, not firm value alone. If debt is stochastic, the default boundary is stochastic too.

3.2 Constant debt: BC-0G

The constant-debt model assumes

$$k_t = k_0 \quad \text{for all } t > 0.$$

Interpretation. This is the common assumption in many structural credit models. It is convenient but counterfactual because nominal debt grows substantially on average.

3.3 Deterministic debt growth: BC

The Black-Cox model with constant debt growth assumes

$$k_t = k_0 + \gamma t.$$

Interpretation. This model allows average debt to grow, but all firms grow debt at the same deterministic rate. It cannot match cross-sectional differences in debt growth across leverage bins.

3.4 Deterministic debt adjustment: CDG

The Collin-Dufresne and Goldstein model assumes

$$dk_t = \lambda(\nu - l_t)dt.$$

If leverage is above target, debt falls; if leverage is below target, debt rises.

Interpretation. CDG introduces mean-reverting leverage and captures long-run target leverage behavior. But because the adjustment is locally deterministic, short-run leverage shocks still come only from asset-value shocks.

3.5 Stochastic debt adjustment: SD

The paper's stochastic-debt model assumes

$$dk_t = \lambda(\nu - l_t)dt + \sigma_k dW_{k,t},$$

where $dW_{k,t}$ is correlated with the asset-value shock dW_t :

$$\text{corr}(dW_t, dW_{k,t}) = \rho.$$

Interpretation. The stochastic debt shock captures short-run financing behavior that is not mechanically implied by firm-value shocks. A negative ρ means bad firm-value shocks tend to coincide with positive debt shocks, matching short-run pecking-order behavior.

3.6 Parameter estimation for debt dynamics

The paper estimates debt-dynamics parameters by matching:

- average historical debt growth by leverage bin and horizon;
- the difference in debt growth between high- and low-equity-return firms.

The objective is a weighted least-squares distance between historical and model-implied growth rates:

$$\min_{\Theta_k} \sum_{i=1}^5 \sum_{T=1}^{10} \left[\left(\frac{D_{i,T}^g - D_T^g(L_i^m, \hat{\Theta}_v^P, \Theta_k)}{SD_{i,T}^D} \right)^2 + \left(\frac{\Delta D_{i,T}^g - \Delta D_T^g(L_i^m, \hat{\Theta}_v^P, \Theta_k)}{SD_{i,T}^{\Delta D}} \right)^2 \right].$$

Interpretation. The fitting targets both unconditional debt growth and conditional debt-growth responses to equity shocks. This is why the SD model is disciplined by debt-policy facts rather than by spread data.

3.7 Leverage volatility

In the CDG and SD models, log leverage has Ornstein-Uhlenbeck dynamics:

$$dl_t = \lambda(\bar{l} - l_t)dt + \sigma_l dW_{l,t}.$$

In the CDG model, $\sigma_l = \sigma$. In the SD model,

$$\sigma_l = \sqrt{\sigma_k^2 + \sigma^2 - 2\rho\sigma_k\sigma}.$$

The paper estimates the ratio of leverage volatility to asset volatility as

$$\omega \equiv \frac{\sigma_l}{\sigma} = 1.6409.$$

Interpretation. Stochastic debt makes leverage much more volatile than assets at short horizons. This is the key quantitative channel for credit spreads.

3.8 Default-boundary calibration

For each model, the default boundary d is chosen to match historical default rates across ratings and maturities:

$$\min_d \sum_{a=AAA}^B \sum_{T=1}^{20} \frac{1}{T} |\pi_{aT}^P(d) - \hat{\pi}_{aT}^P|.$$

Interpretation. Since each model has different leverage dynamics, each model requires a different default boundary to match the same historical default rates.

3.9 Credit spread pricing

For a zero-coupon bond with maturity T , recovery R , risk-free rate r , and risk-neutral default probability $\pi^Q(T)$, the model spread is

$$s = y - r = -\frac{1}{T} \log [1 - (1 - R)\pi^Q(T)].$$

The paper uses a senior-unsecured bond recovery rate of 33.48%.

Interpretation. Once the default boundary is calibrated to historical defaults, the model's ability to match observed spreads depends on its risk-neutral default probabilities and their sensitivity to leverage, volatility, and maturity.

3.10 Distance-to-default intuition

A useful approximation is

$$DD_T = \frac{\ln(V_0/K_0) - \ln(d) + (\mu_l - \frac{1}{2}\sigma_l^2)T}{\sigma_l\sqrt{T}}.$$

The sensitivity of distance-to-default to leverage is roughly

$$-\frac{1}{\sigma_l\sqrt{T}}.$$

Interpretation. A higher σ_l lowers the sensitivity of default risk to observed leverage. This explains why the SD model predicts higher spreads for low-leverage firms and lower spreads for very high-leverage firms relative to standard diffusion models.

3.11 Stochastic-volatility/jump extension

For the Walmart CDS exercise, the paper studies a model with stochastic asset volatility, jumps in firm value, and stochastic debt:

$$\begin{aligned}\frac{dV_t}{V_t} &= (r - \delta - \xi\eta^Q)dt + \sqrt{A_t}dW_{1,t} + dJ_t^Q, \\ dA_t &= \lambda_A(\nu_A - A_t)dt + \sigma_A dW_{2,t}, \\ dk_t &= \lambda(\nu - l_t)dt + \sigma_k dW_{3,t}.\end{aligned}$$

Interpretation. This exercise asks whether stochastic debt matters even when the asset-value process is richer than a simple diffusion. The answer is yes: adding stochastic debt sharply improves the fit to Walmart CDS curves.

4 Identification and empirical design

4.1 What identifies the debt-dynamics facts

- The main debt-policy evidence compares firms within similar initial leverage bins.
- Within each leverage bin, firms are split by future equity returns.
- The comparison asks how debt evolves for firms hit by high versus low equity-return shocks.

Interpretation. The leverage-bin design reduces the concern that high- and low-return firms simply start with different leverage. The short-run and long-run patterns then reveal how debt policy responds to shocks.

4.2 Why the short-run result matters

- A pure target-leverage model predicts that firms hit by negative shocks should reduce debt or increase it less.
- The data show the opposite at short horizons: low-equity-return firms often issue more debt.
- This pattern is consistent with pecking-order or liquidity behavior: firms hit by shocks borrow to absorb the shock.

Interpretation. This is the empirical fact that the CDG model cannot match and that motivates adding stochastic debt shocks.

4.3 Why the long-run result matters

- After the shock window, high-return firms increase debt more than low-return firms.
- Low-leverage firms increase debt more than high-leverage firms.
- Highly levered firms tend to reduce debt.

Interpretation. These patterns are consistent with target leverage and motivate mean reversion in the debt process.

4.4 Pricing identification

- The models are calibrated to default rates and debt dynamics.
- Observed bond spreads are not used to estimate the main model parameters.
- The spread comparisons are therefore out-of-sample pricing tests.

Interpretation. This design makes the pricing evidence more informative. If the SD model fits spreads better, it is not because it was directly fitted to those spreads.

4.5 Robustness checks

The paper checks that the key conclusions survive:

- alternative sample periods;
- survivorship restrictions;

- exact matching on firm size, leverage, and cash;
- Treasury yields rather than swap rates;
- CDS spreads rather than bond spreads;
- market values of debt rather than book values;
- alternative estimation of debt-asset correlation.

Interpretation. The robustness evidence is especially important because the paper connects two empirical objects: debt dynamics and credit-spread pricing.

4.6 What the paper cannot fully prove

- The debt response to equity-return shocks is not a randomized experiment.
- Book debt is an imperfect proxy for market debt, especially near distress.
- Asset volatility and payout rates must be estimated.
- The stochastic-volatility/jump extension is implemented only for Walmart, so it is exploratory rather than a full-sample test.

Interpretation. The paper's strongest contribution is not a single clean natural experiment. It is the coherent alignment of debt-dynamics facts, structural-model implications, and out-of-sample spread performance.

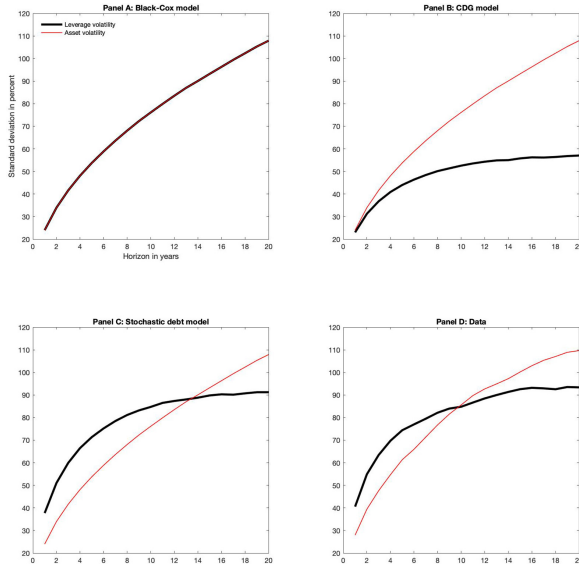
5 Tables and figures

5.1 Figure 1 and Figure 2: why stochastic debt is needed

Read:

- Figure 1 compares leverage volatility and asset volatility across models and data.
- In the Black-Cox model, leverage volatility and asset volatility coincide.
- In the CDG model, leverage volatility is lower than asset volatility at longer horizons because leverage mean reverts.
- In the SD model, leverage volatility is higher than asset volatility at short horizons but eventually mean reverts.
- The empirical pattern in Panel D resembles the SD model: high short-run leverage volatility, then slower long-run growth.
- Figure 2 shows that low-leverage firms grow debt much more than high-leverage firms.

Economic message. The empirical relation between asset volatility and leverage volatility cannot be captured by constant-debt or deterministic-adjustment models. The SD model matches the short-run extra volatility and long-run mean reversion.



1. Leverage and asset volatility: This graph shows volatility of leverage - $\sqrt{\text{Var}(\log(\frac{L_{t+T}}{L_t}))}$ - and volatility of asset value - $\sqrt{\text{Var}(\log(\frac{V_{t+T}}{V_t}))}$ - as a function of horizon $T - t$ in years. In the Black-Cox model log-firm value is given as $dv_t = (\mu - \delta - \frac{\sigma^2}{2})dt + \sigma dW_t$ and the only source of volatility in leverage is the volatility of the firm value. Panel A shows the volatility for different horizons, where we have used average parameters from the empirical section, $\sigma = 0.24$, $\mu = 0.1028$, and $\delta = 0$.

Figure 1. Leverage and asset volatility.

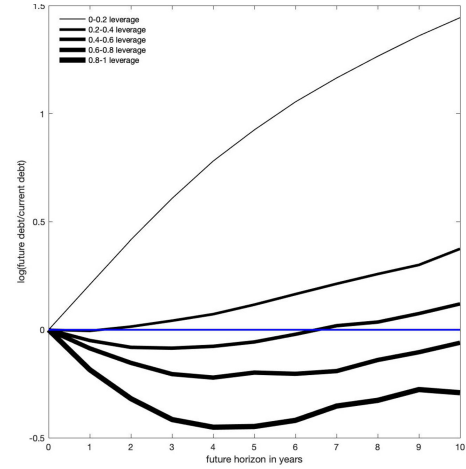


Fig. 2. Future debt growth as a function of initial leverage. For firm i , year t , and horizons 1,...,10 years, we calculate $\log(\frac{D_{i,t+T}}{D_{i,t}})$ where $D_{i,t}$ is the nominal level of debt for firm i in year t and T is the horizon in years. The figure shows the average ratio for different initial leverage ratios and future horizons. The data is from CRSP/Compustat and the sample period is 1988–2017.

Figure 2. Future debt growth by initial leverage.

5.2 Table 1 and Table 2: debt growth and equity-return shocks

Read:

- Table 1 shows that nominal debt grows substantially: for all firms, the increase is 9% after one year, 60% after five years, and 136% after ten years.
- Firms with leverage below 20% increase debt by 23%, 152%, and 324% over one, five, and ten years.
- Firms with leverage above 80% reduce debt by 17%, 36%, and 25% over the same horizons.
- Table 2 conditions on future three-year equity returns.
- Low-return firms often issue more debt at the one- and two-year horizons.
- At longer horizons, high-return firms increase debt more, consistent with gradual movement toward target leverage.

Economic message. The short-run response looks like pecking-order or liquidity behavior; the long-run response looks like target leverage. A useful credit-risk model must contain both.

Table 1

Future debt relative to current debt. For firm i , year t , and horizons 1,...,10 years, we calculate $\log(\frac{D_{i,t+T}}{D_{i,t}})$ where $D_{i,t}$ is the nominal level of debt for firm i in year t and T is the horizon in years. The table shows the average log-ratio for different initial leverage ratios and future horizons. The final columns of the table shows for selected horizons the growth rate calculated as $\exp(\log\text{-growth rate}) - 1$. Standard errors clustered at firm level is in parentheses and the number of observations in brackets. The data is from CRSP/Compustat and the sample period is 1988–2017.

Horizon (years)	Log-growth										Growth (in percent)		
	1	2	3	4	5	6	7	8	9	10	1	5	10
Leverage 0–1	0.09 (0.00) [94,138]	0.18 (0.00) [81,591]	0.28 (0.01) [71,423]	0.38 (0.01) [63,079]	0.47 (0.01) [56,081]	0.55 (0.01) [49,990]	0.63 (0.01) [44,573]	0.70 (0.02) [39,795]	0.78 (0.02) [35,421]	0.86 (0.02) [31,522]	9	60	136
Leverage 0–0.2	0.21 (0.00) [47,690]	0.42 (0.01) [41,374]	0.61 (0.01) [36,285]	0.78 (0.01) [32,046]	0.92 (0.02) [28,509]	1.05 (0.02) [25,494]	1.16 (0.02) [22,753]	1.27 (0.03) [20,375]	1.36 (0.03) [18,294]	1.44 (0.03) [16,459]	23	152	324
Leverage 0.2–0.4	–0.01 (0.00) [23,453]	0.01 (0.01) [20,624]	0.04 (0.01) [18,259]	0.07 (0.01) [16,214]	0.12 (0.01) [14,478]	0.17 (0.02) [12,889]	0.21 (0.02) [11,522]	0.26 (0.02) [10,281]	0.30 (0.03) [9,148]	0.37 (0.03) [8,160]	–1	12	45
Leverage 0.4–0.6	–0.05 (0.00) [13,055]	–0.08 (0.01) [11,410]	–0.09 (0.01) [9,910]	–0.08 (0.01) [8,729]	–0.06 (0.02) [7,699]	–0.02 (0.02) [6,837]	0.02 (0.02) [6,076]	0.04 (0.03) [5,450]	0.08 (0.03) [4,810]	0.12 (0.03) [4,226]	–5	–5	13
Leverage 0.6–0.8	–0.09 (0.01) [7,238]	–0.15 (0.01) [6,025]	–0.21 (0.01) [5,156]	–0.22 (0.02) [4,516]	–0.20 (0.02) [4,016]	–0.20 (0.03) [3,567]	–0.19 (0.03) [3,165]	–0.14 (0.03) [2,766]	–0.10 (0.04) [2,397]	–0.06 (0.05) [2,037]	–8	–18	–6
Leverage 0.8–1	–0.18 (0.01) [2,702]	–0.32 (0.02) [2,158]	–0.42 (0.03) [1,813]	–0.45 (0.04) [1,574]	–0.45 (0.04) [1,379]	–0.42 (0.05) [1,203]	–0.35 (0.06) [1,057]	–0.33 (0.07) [923]	–0.28 (0.09) [772]	–0.29 (0.10) [640]	–17	–36	–25

Table 1. Future debt relative to current debt.

Table 2

Future debt relative to current debt conditional on future three-year equity returns. For firm i , year t , and horizons 1,...,10 years, we calculate $\log\left(\frac{D_{i,t+T}}{D_{i,t}}\right)$ where $D_{i,t}$ is the nominal level of debt for firm i in year t and T is the horizon in years. For each firm-year in the sample where the initial leverage ratio at time t of the firm is in a certain interval, we calculate the future three-year equity return between t and $t+3$ and label firms with a return higher (lower) than the (within this leverage group) median between t and $t+3$ 'High (Low) future equity return' firms. The table reports the average log-ratio for high and low future equity return firms as well as the difference. The final column shows the average initial leverage ratio. In parentheses are standard errors, clustered at the firm level, of the differences and '**' indicate significance at 95% level and '***' at 99% level. The data is from CRSP/Compustat and the sample period is 1988–2017.

Horizon (years)	1	2	3	4	5	6	7	8	9	10	Average leverage
Leverage 0–0.2											
High equity return	0.21	0.43	0.61	0.85	1.03	1.18	1.31	1.40	1.48	1.56	0.07
Low equity return	0.31	0.53	0.60	0.69	0.75	0.82	0.88	0.96	1.06	1.14	0.08
Difference	-0.10**	-0.09**	0.01	0.16**	0.28**	0.36**	0.43**	0.44**	0.41**	0.42**	-0.01**
	(0.01)	(0.02)	(0.02)	(0.03)	(0.03)	(0.04)	(0.04)	(0.04)	(0.05)	(0.05)	(0.00)
Leverage 0.2–0.4											
High equity return	-0.00	0.02	0.07	0.18	0.27	0.35	0.43	0.51	0.55	0.64	0.29
Low equity return	0.04	0.05	0.02	-0.02	-0.03	-0.03	-0.02	-0.01	0.02	0.06	0.29
Difference	-0.04**	-0.03*	0.05**	0.19**	0.30**	0.37**	0.45**	0.52**	0.53**	0.58**	0.00
	(0.01)	(0.01)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)	(0.04)	(0.05)	(0.00)
Leverage 0.4–0.6											
High equity return	-0.05	-0.06	-0.05	0.04	0.10	0.18	0.24	0.29	0.35	0.38	0.49
Low equity return	-0.02	-0.05	-0.12	-0.17	-0.20	-0.21	-0.20	-0.23	-0.21	-0.18	0.49
Difference	-0.02*	-0.00	0.06**	0.21**	0.31**	0.39**	0.44**	0.52**	0.56**	0.57**	-0.00
	(0.01)	(0.02)	(0.02)	(0.03)	(0.03)	(0.04)	(0.04)	(0.05)	(0.05)	(0.06)	(0.00)
Leverage 0.6–0.8											
High equity return	-0.08	-0.13	-0.17	-0.14	-0.05	-0.03	0.02	0.06	0.07	0.12	0.69
Low equity return	-0.07	-0.14	-0.24	-0.28	-0.31	-0.36	-0.40	-0.32	-0.29	-0.25	0.69
Difference	-0.01	0.01	0.07**	0.13**	0.26**	0.33**	0.42**	0.38**	0.36**	0.37**	-0.00
	(0.01)	(0.02)	(0.03)	(0.03)	(0.04)	(0.05)	(0.05)	(0.06)	(0.07)	(0.08)	(0.00)
Leverage 0.8–1											
High equity return	-0.16	-0.28	-0.38	-0.33	-0.28	-0.21	-0.11	-0.06	0.02	0.03	0.87
Low equity return	-0.19	-0.32	-0.45	-0.55	-0.59	-0.64	-0.61	-0.64	-0.68	-0.72	0.88
Difference	0.02	0.04	0.06	0.22**	0.31**	0.43**	0.50**	0.58**	0.71**	0.74**	-0.01**
	(0.03)	(0.04)	(0.05)	(0.07)	(0.08)	(0.10)	(0.11)	(0.12)	(0.15)	(0.19)	(0.00)

Table 2. Future debt relative to current debt, conditional on future three-year equity returns.

5.3 Figure 3 and Figure 4: short-run borrowing after bad shocks

Read:

- Figure 3 shows the cumulative debt-growth path for high- and low-equity-return firms.
- For leverage bins below 80%, low-return firms generally increase debt more in the first one to two years.
- After the three-year return window, high-return firms increase debt more.
- Figure 4 decomposes this behavior into short-term debt, long-term debt, cash, total debt minus cash, and leverage.
- The short-run difference is strongest in short-term debt.
- Low-return firms do not simply hoard cash; leverage rises sharply after negative shocks.

Economic message. Negative shocks push firms closer to default not only because asset values fall, but also because debt can rise. This is exactly the channel missing from deterministic debt-adjustment models.

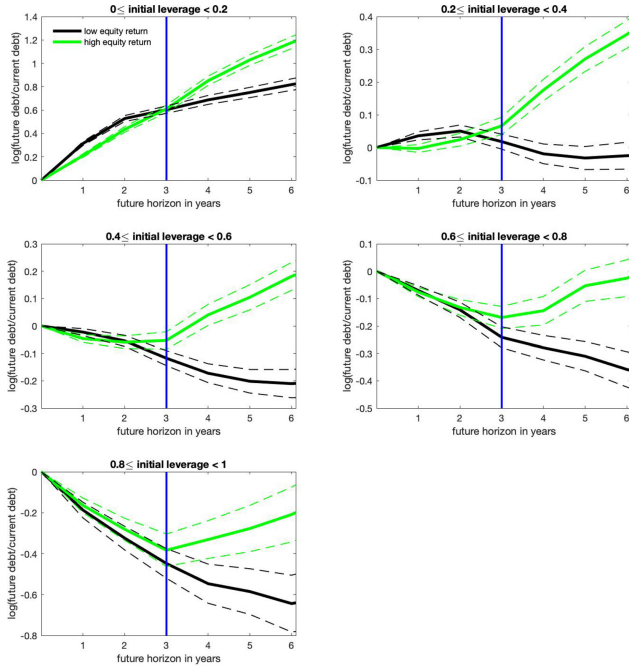


Figure 3. Debt growth conditional on future equity returns. For firm i , year t , and horizons 1,...,20, we calculate $\log\left(\frac{D_{i,t+T}}{D_{i,t}}\right)$ where $D_{i,t}$ is the level of debt for firm i in year t and T is the horizon in years. For each firm-year in the sample where the initial leverage ratio at time t of the firm is in the interval, we calculate the future three-year equity return between t and $t+3$ and label firms with a return higher (lower) than the (within this leverage group) median between t and $t+3$ 'High (Low) future equity return' firms. The figure shows the average log-ratio for high and low future equity return firms. The dashed lines mark 95% confidence levels based on standard errors clustered at the firm level. The data is from CRSP/Compustat and the sample period is 1988–2017.

Figure 3. Debt growth conditional on future equity returns.

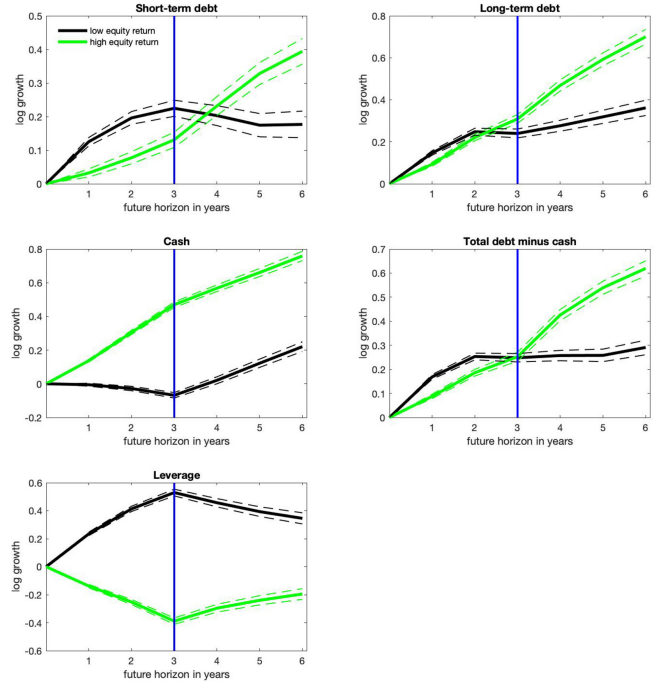


Figure 4. Short-term debt, long-term debt, cash, and leverage. For firm i , year t , and horizons 1,...,20, we calculate $\log\left(\frac{B_{i,t+T}}{B_{i,t}}\right)$ where $B_{i,t}$ is the variable of interest for firm i in year t and T is the horizon in years. For each firm-year in the sample where the initial leverage ratio at time t of the firm is in the interval, we calculate the future three-year equity return between t and $t+3$ and label firms with a return higher (lower) than the (within this leverage group) median between t and $t+3$ 'High (Low) future equity return' firms. The figure shows the average log-ratio for high and low future equity return firms. The dashed lines mark 95% confidence levels based on standard errors clustered at the firm level. The data is from CRSP/Compustat and the sample period is 1988–2017.

Figure 4. Short-term debt, long-term debt, cash, and leverage.

5.4 Figure 5 and Figure 6: model fit to debt dynamics

Read:

- Figure 5 compares observed debt growth with fitted values from BC-0G, BC, CDG, and SD.
- BC-0G misses the average growth of debt.
- BC allows deterministic debt growth but cannot match cross-sectional differences across leverage bins.
- CDG and SD fit unconditional debt growth much better because both have mean-reverting leverage.
- Figure 6 compares the difference in debt growth between high- and low-equity-return firms.
- CDG predicts that high-return firms always issue more debt.
- SD matches the short-run negative difference and the long-run positive difference.

Economic message. CDG captures target leverage but not shock-driven borrowing. SD preserves target leverage while adding stochastic short-run debt shocks.

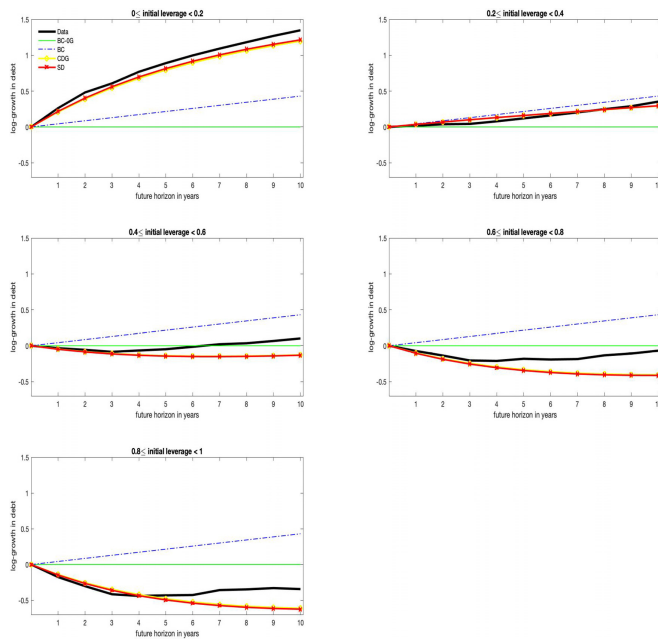


Figure 5. Model fit to future debt growth. For firm i , year t , and horizons 1,...,10 years, we calculate $\log\left(\frac{D_{i,t+T}}{D_{i,t}}\right)$ where $D_{i,t}$ is the nominal level of debt for firm i in year t and T is the horizon in years. 'Data' shows the average log-ratio for different initial leverage ratios and future horizons in the data. The figure also shows fitted values from structural models. 'BC-0G' refers to the Black-Cox model with zero growth in debt. 'BC' refers to the Black-Cox model where all firms have a common long-run target leverage. 'SD' refers to a model where the firm adjusts the level of debt such that a long-run leverage, common to all firms, is targeted, and the level of debt is subject to random shocks. The data is from CRSP/Compustat and the sample period is 1988–2017.

Figure 5. Model fit to future debt growth.

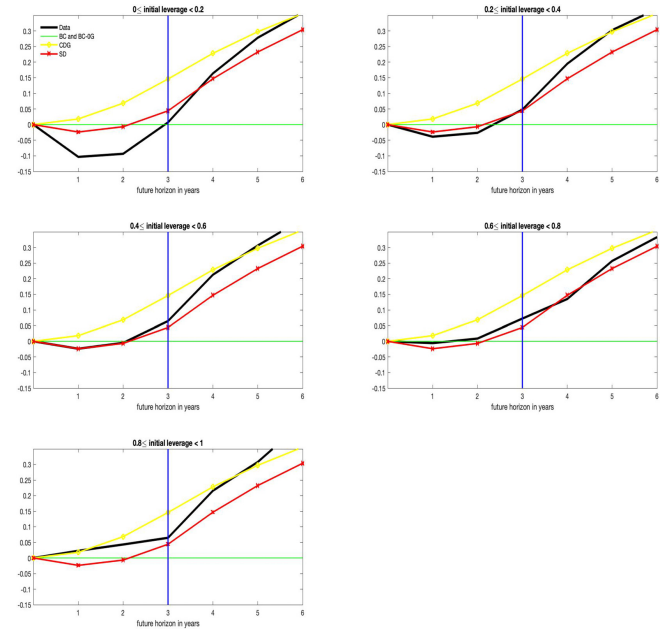


Figure 6. High-return minus low-return debt growth. For firm i , year t , and horizons 1,...,20 years, we calculate $\log\left(\frac{D_{i,t+T}}{D_{i,t}}\right)$ where $D_{i,t}$ is the nominal level of debt for firm i in year t and T is the horizon in years. For each firm-year where the initial leverage ratio at time t of the firm is in a certain interval, we calculate the future three-year equity return between t and $t+3$ for firms with a return higher (lower) than the (within this leverage group) median 'High (Low) future equity return' firms. The figure shows 1 values for the different models and the values from the data of the difference in the average increase in log-debt for high and low future returns. 'BC-0G' refers to the Black-Cox model with zero growth in debt. 'BC' refers to the Black-Cox model where all firms have a common long-run target leverage. 'SD' refers to a model where the firm adjusts the level of debt such that a long-run leverage, common to all firms, is targeted, and the level of debt is subject to random shocks. The data is from CRSP/Compustat and the sample period is 1988–2017.

Figure 6. High-return minus low-return debt growth.

5.5 Table 3 and Table 4: bond and firm samples

Read:

- Table 3 reports 101,059 bond observations and 3,129 bonds with 0–20 year maturity.
- Most observations are investment grade, especially A and BBB.
- The sample includes 26,798 short-maturity observations, 52,046 medium-maturity observations, and 22,215 long-maturity observations.
- Table 4 shows that leverage rises strongly as ratings deteriorate: average leverage is 0.12 for AAA, 0.35 for BBB, 0.56 for B, and 0.69 for C.
- Equity volatility also rises with weaker ratings, from 0.24 for AAA to 0.61 for C.
- Asset volatility is much more stable, around 0.22–0.27 across ratings.

Economic message. Poorer credit ratings are associated mainly with higher leverage and equity volatility. Asset volatility varies less than equity volatility because leverage mechanically amplifies equity risk.

Table 3

Bond summary statistics. The sample consists of noncallable bonds with fixed coupons issued by industrial firms. This table shows summary statistics for the data set. The data sample cover the period 1988Q2-2018Q1. 'Number of bonds' is the number of bonds that appear (in a particular rating and maturity range) at some point in the sample period. For each quote we calculate the bond's time since issuance and 'Age' is the average time since issuance across all quotes. 'Coupon' is the average bond coupon across all quotes. 'Amount outstanding' is the average outstanding amount of a bond issue across all quotes. 'Time-to-maturity' is the average time until the bond matures across all quotes.

	AAA	AA	A	BBB	BB	B	C	all
0–20-year bond maturity								
Number of bonds	66	322	1486	1422	563	410	152	3129
Age	4.99	4.68	6.1	7.24	6.79	7.82	11.1	6.59
Coupon	5.6	6.23	6.99	7.61	7.58	7.46	7.73	7.18
Amount outstanding (\$mm)	610	520	296	324	274	282	275	333
Time-to-maturity	6.87	6.45	6.69	7.11	6.78	6.16	7.90	6.82
Number of observations	2617	10,667	36,144	31,172	11,907	5679	2873	101,059
0–3-year bond maturity								
Number of bonds	26	144	765	730	262	238	61	1884
Age	4.72	5.78	6.95	8.25	7.05	7.2	9.58	7.29
Coupon	3.94	4.73	5.82	6.62	6.92	6.63	7.44	6.15
Amount outstanding (\$mm)	707	707	342	361	293	297	304	382
Time-to-maturity	1.59	1.43	1.45	1.39	1.51	1.47	1.61	1.44
Number of observations	513	2706	9668	8228	3072	1882	729	26,798
3–10-year bond maturity								
Number of bonds	52	230	1025	864	330	184	80	2219
Age	3.43	3.77	4.73	5.78	5.41	6.78	8.74	5.16
Coupon	5.81	6.47	7.19	7.86	7.72	7.84	7.54	7.37
Amount outstanding (\$mm)	667	488	291	326	280	292	258	334
Time-to-maturity	6.17	6.23	6.11	6.34	6.02	5.73	5.74	6.15
Number of observations	1605	6121	18,967	15,385	6263	2608	1097	52,046
10–20-year bond maturity								
Number of bonds	16	80	293	347	129	77	51	745
Age	10.3	6.09	8.48	9.11	9.81	11	14.5	9.12
Coupon	6.65	7.6	8	8.15	8.04	7.93	8.13	8
Amount outstanding (\$mm)	329	350	252	281	235	235	274	270
Time-to-maturity	14.55	14.56	14.92	14.91	14.93	14.52	14.54	14.84
Number of observations	499	1840	7509	7559	2572	1189	1047	22,215

Table 3. Bond summary statistics.

Table 4

Firm summary statistics. For each bond yield observation, the leverage ratio, equity volatility, asset volatility, and payout ratio are calculated for the issuing firm on the day of the observation. Leverage ratio is the ratio of the book value of debt to the market value of equity plus the book value of debt. Equity volatility is the annualized volatility of daily equity returns from the last three years. Asset volatility is the unlevered equity volatility, calculated as explained in the text. Payout ratio is yearly interest payments plus dividends plus share repurchases divided by firm value. Note that since some firms change rating in our sample period, the sum of firms over rating classes is greater than 'all'. Firm variables are computed using data from CRSP and Compustat.

	#firms	Mean	10th	25th	Median	75th	90th
Leverage ratio							
AAA	15	0.12	0.03	0.05	0.07	0.14	0.25
AA	68	0.17	0.07	0.11	0.17	0.20	0.24
A	231	0.28	0.11	0.17	0.25	0.36	0.50
BBB	353	0.35	0.15	0.23	0.32	0.44	0.55
BB	208	0.44	0.20	0.32	0.42	0.56	0.68
B	122	0.56	0.26	0.38	0.58	0.71	0.87
C	50	0.69	0.41	0.56	0.72	0.87	0.95
all	571	0.33	0.11	0.18	0.29	0.44	0.60
Equity volatility							
AAA	15	0.24	0.16	0.19	0.23	0.29	0.36
AA	68	0.25	0.16	0.20	0.24	0.29	0.38
A	231	0.31	0.19	0.24	0.29	0.36	0.41
BBB	353	0.33	0.21	0.25	0.31	0.39	0.47
BB	208	0.44	0.27	0.31	0.40	0.50	0.62
B	122	0.55	0.28	0.35	0.48	0.68	0.85
C	50	0.61	0.36	0.44	0.56	0.74	0.98
all	571	0.34	0.20	0.24	0.31	0.40	0.50
Asset volatility							
AAA	15	0.22	0.17	0.20	0.20	0.24	0.25
AA	68	0.22	0.20	0.20	0.22	0.24	0.25
A	231	0.24	0.18	0.20	0.23	0.25	0.30
BBB	353	0.24	0.16	0.20	0.22	0.26	0.33
BB	208	0.26	0.17	0.21	0.26	0.31	0.34
B	122	0.27	0.17	0.22	0.25	0.30	0.37
C	50	0.26	0.11	0.19	0.23	0.31	0.40
all	571	0.24	0.17	0.20	0.23	0.26	0.32
Payout ratio							
AAA	15	0.042	0.012	0.021	0.044	0.059	0.073
AA	68	0.037	0.010	0.018	0.034	0.050	0.065
A	231	0.041	0.017	0.024	0.037	0.052	0.071
BBB	353	0.047	0.018	0.028	0.042	0.059	0.085
BB	208	0.046	0.018	0.027	0.040	0.055	0.076
B	122	0.049	0.023	0.033	0.044	0.060	0.079
C	50	0.051	0.027	0.039	0.049	0.059	0.073
all	571	0.044	0.016	0.025	0.040	0.055	0.076

Table 4. Firm summary statistics.

5.6 Table 5 and Figure 7: default boundaries and leverage sensitivity

Read:

- Table 5 gives default-boundary estimates for each model.
- The BC-0G boundary is 0.8614; the BC boundary is 0.7315.
- The CDG boundary is very high at 0.9978, while SD is much lower at 0.7322 and SD-FL is 0.6707.
- The lower SD boundary reflects higher leverage volatility.
- Figure 7 focuses on short-term investment-grade spreads across leverage quintiles.
- For low-to-moderate leverage quintiles, SD and SD-FL are generally closer to actual spreads than the other models.

Economic message. Higher short-run leverage volatility changes the mapping from leverage to spreads. After calibration to default rates, SD dampens the extreme sensitivity to leverage that standard diffusion models generate.

Table 5

Estimates of the default boundary. For each of the models we estimate the default boundary – the fraction of the total face value of debt at which the firm defaults – by minimizing the distance between average model-implied default probabilities and historical average default rates for a cross-section of horizons 1,...,20 years and ratings AAA, AA, A, BBB, BB, B. 'BC-OG' to the Black-Cox model with zero growth in debt. 'BC' refers to the Black-Cox model. 'CDG' refers to the *Collin-Dufresne and Goldstein (2001)* model where all firms have a common long-run target leverage. 'SD' refers to a model where the firm adjusts the level of debt such that a long-run leverage, common to all firms, is targeted, and the level of debt is subject to random shocks. 'FL' refers to models where the long-run target leverage is firm specific and calculated as the historical average firm leverage. The historical average default rates are based on defaults of U.S. industrial firms in the period 1970–2017.

Model	Default boundary estimate ($\hat{d}$)
BC-OG	0.8614
BC	0.7315
CDG	0.9978
CDG-FL	0.9103
SD	0.7322
SD-FL	0.6707

Table 5. Default-boundary estimates.

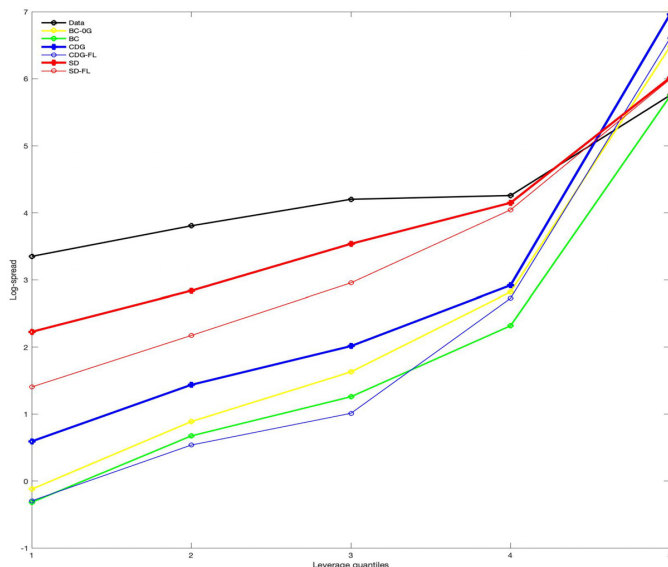


Figure 7. Short-term investment-grade spreads by leverage quintile. For bonds with an investment grade rating and we calculate the average actual spread (in bps) of all bond-month observations where the issuing firm has a leverage quantile, ..., fifth-lowest quantile and compute the log of the average spread. For each leverage quintile, the figure shows the data and for each of the models. The sample period is 2002:07–2018:03.

Figure 7. Short-term investment-grade spreads by leverage quintile.

5.7 Table 6 and Table 7: average spreads and time-series pricing errors

Read:

- Table 6 shows average actual and model-implied spreads across maturity buckets.
- For investment-grade bonds, actual average spreads are 84 bps overall, 77 bps short, 72 bps medium, and 121 bps long.
- SD-FL is close overall for investment-grade bonds: 86 bps overall and 111 bps long.
- For speculative-grade bonds, all models tend to underpredict long-maturity spreads.
- Table 7 reports monthly absolute pricing errors.
- Across all bonds and maturities, SD has the lowest average error at 52 bps; SD-FL is next at 57 bps.
- For investment-grade short-maturity bonds, SD has an error of 36 bps and SD-FL 38 bps, much lower than BC-OG, BC, and CDG.

Economic message. The main pricing improvement comes from short-maturity investment-grade bonds, historically the hardest category for simple diffusion structural models.

Table 6

Actual and model credit spreads. For every model we imply out the default boundary by minimizing the difference between model-implied and historical default rates. This table shows average actual and model-implied corporate bond yield spreads. Spreads are grouped according to remaining bond maturity at the spread observation date. 'Actual spread' is the actual spread to the swap rate. 'BC-OG' to the Black-Cox model with zero growth in debt. 'BC' refers to the Black-Cox model. 'CDG' refers to the [Collin-Dufresne and Goldstein \(2001\)](#) model where all firms have a common long-run target leverage. 'SD' refers to a model where the firm adjusts the level of debt such that a long-run leverage, common to all firms, is targeted, and the level of debt is subject to random shocks. 'FL' refers to models where the long-run target leverage is firm specific and calculated as the historical average firm leverage. 'Inv' includes bonds rated AAA, AA, A, and BBB, while 'Spec' includes bonds rated BB, B, and C. Confidence bands are simulation-based following [Feldhütter and Schaefer \(2018\)](#) and are at the 95% level. * implies significance at the 5% level and ** at the 1% level. The sample period is 1988–2018 for spread of bonds with a maturity more than three years and 2002–2018 for bonds with a maturity less than three years.

		All	Short	Medium	Long
Inv	Actual spread	84	77	72	121
	BC-OG	88 (49;132)	73 (22;162)	82 (46;112)	122 (91;142)
	BC	99 (64;128)	36* (12;66)	93 (55;123)	189** (149;214)
	CDG	73 (25;123)	117 (21;217)	51 (22;84)	72** (38;103)
	CDG-FL	60 (21;109)	81 (18;183)	53 (21;86)	52*** (24;76)
	SD	107 (62;144)	67 (28;110)	107 (63;143)	153 (103;189)
	SD-FL	86 (48;119)	59 (25;94)	89 (50;122)	111 (70;140)
	Spec Actual spread	413	409	387	470
	BC-OG	392 (223;578)	447 (162;873)	401 (250;518)	308*** (240;354)
	BC	363 (238;468)	248 (103;415)	412 (269;514)	398*** (333;435)
	CDG	345 (119;607)	641 (152;1208)	259 (116;426)	170*** (86;270)
	CDG-FL	385 (166;621)	502 (147;1008)	371 (179;528)	275*** (159;353)
	SD	349 (210;466)	423 (208;635)	353 (227;448)	251*** (177;302)
	SD-FL	400 (249;518)	408 (209;594)	428 (277;536)	334*** (239;392)

Table 6. Actual and model credit spreads.

Table 7

Pricing errors of monthly credit spreads. This table shows how well structural models match average monthly credit spreads. For a given rating r and maturity m , we find all bonds at the end of a given month t that have this rating and maturity, calculate the average actual credit spread (in basis points) to the swap rate, s_{rmt}^a , and do this for all months in the sample. For each model, we likewise calculate a time series of monthly average model credit spread (in basis points) $s_{rmt}^M, \dots, s_{rmt}^M$. This table shows the average absolute pricing error $1/T \sum_{t=1}^T |s_{rmt}^a - s_{rmt}^M|$. 'Short' includes bond maturities in the range 0–3 years, 'Medium' 3–10 years, and 'Long' 10–20 years. 'BC-OG' refers to the Black-Cox model with zero growth in debt. 'BC' refers to the Black-Cox model. 'CDG' refers to the [Collin-Dufresne and Goldstein \(2001\)](#) model where all firms have a common long-run target leverage. 'SD' refers to a model where the firm adjusts the level of debt such that a long-run leverage, common to all firms, is targeted, and the level of debt is subject to random shocks. 'FL' refers to models where the long-run target leverage is firm specific and calculated as the historical average firm leverage. 'Inv' includes bonds rated AAA, AA, A, and BBB, while 'Spec' includes bonds rated BB, B, and C. The sample period for 'Short' is 2002:07–2018:03 while it is 1988:03–2018:03 for 'Medium' and 'Long'.

		Average	Short	Medium	Long
All	BC-OG	63	109	40	42
	BC	67	92	46	62
	CDG	119	204	53	99
	CDG-FL	87	127	41	92
	SD	52	65	50	42
Inv	SD-FL	57	67	50	53
	BC-OG	44	72	31	29
	BC	55	53	37	76
	CDG	75	136	34	53
	CDG-FL	62	83	33	72
Spec	SD	45	36	51	48
	SD-FL	39	38	42	37
	BC-OG	194	247	169	166
	BC	181	241	173	130
	CDG	285	357	223	277
	CDG-FL	218	282	177	196
	SD	185	180	165	211
	SD-FL	179	185	185	166

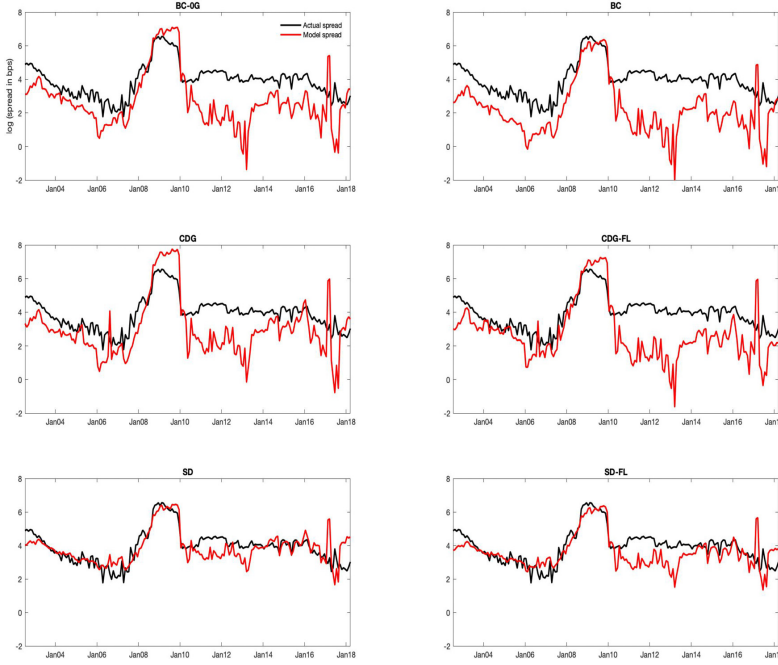
Table 7. Pricing errors of monthly credit spreads.

5.8 Figure 8 and Table 8: time-series fit and spread volatility

Read:

- Figure 8 plots monthly short-term investment-grade log spreads.
- BC-OG, CDG, and CDG-FL tend to underpredict normal-period spreads and become too sensitive during the crisis.
- SD and SD-FL track the data more closely across normal and crisis periods.
- Table 8 shows the standard deviation of monthly spread changes.
- For all short-maturity bonds, the data volatility is 23 bps; SD is 24 bps and SD-FL is 23 bps.
- CDG and CDG-FL are too volatile for short-maturity bonds, at 61 bps and 41 bps.

Economic message. SD improves not only average pricing errors but also the volatility of spread changes. This supports the idea that standard models make spreads too sensitive to leverage movements.



Actual and model short term investment grade credit spreads. For the investment grade rating, bond maturity less than three years, and each month in the sample, we find all bonds at the end of a given month t that have this rating and maturity, and calculate the average actual credit spread $s_{t,m}^A$. For each model, we likewise calculate a time series of monthly average model credit spread (in basis points) $s_{t,m}^M, \dots, s_{t,m}^M$. This table shows the standard deviation, in basis points, of monthly changes in the average credit spread. 'Short' includes bond maturities in the range 0–3 years, 'Medium' 3–10 years, and 'Long' 10–20 years. 'BC-OG' refers to the Black-Cox model with zero growth in debt. 'BC' refers to the Black-Cox model. 'CDG' refers to the Collin-Dufresne and Goldstein (2001) model where all firms have a common long-run target leverage. 'SD' refers to a model where the firm adjusts the level of debt such that a long-run leverage, common to all firms, is targeted, and the level of debt is subject to random shocks. 'FL' refers to models where the long-run target leverage is firm specific and calculated as the historical average firm leverage. 'Inv' includes bonds rated AAA, AA, A, and BBB, while 'Spec' includes bonds rated BB, B, and C. The sample period for 'Short' is 2002:07–2018:03 while it is 1988:03–2018:03 for 'Medium' and 'Long'.

Figure 8. Monthly short-term investment-grade spreads.

Table 8

Volatility of monthly credit spreads. This table shows how well structural models match the volatility of monthly credit spreads. For a given rating r and maturity m , we find all bonds at the end of a given month t that have this rating and maturity, calculate the average actual credit spread (in basis points) to the swap rate, $s_{t,m}^A$, and do this for all months in the sample. For each model, we likewise calculate a time series of monthly average model credit spread (in basis points) $s_{t,m}^M, \dots, s_{t,m}^M$. This table shows the standard deviation, in basis points, of monthly changes in the average credit spread. 'Short' includes bond maturities in the range 0–3 years, 'Medium' 3–10 years, and 'Long' 10–20 years. 'BC-OG' refers to the Black-Cox model with zero growth in debt. 'BC' refers to the Black-Cox model. 'CDG' refers to the Collin-Dufresne and Goldstein (2001) model where all firms have a common long-run target leverage. 'SD' refers to a model where the firm adjusts the level of debt such that a long-run leverage, common to all firms, is targeted, and the level of debt is subject to random shocks. 'FL' refers to models where the long-run target leverage is firm specific and calculated as the historical average firm leverage. 'Inv' includes bonds rated AAA, AA, A, and BBB, while 'Spec' includes bonds rated BB, B, and C. The sample period for 'Short' is 2002:07–2018:03 while it is 1988:03–2018:03 for 'Medium' and 'Long'.

		Average	Short	Medium	Long
All	Data	20	23	18	19
	BC-OG	22	34	17	14
	BC	18	18	18	17
	CDG	27	61	12	9
	CDG-FL	23	41	16	12
Inv	SD	15	24	12	10
	SD-FL	16	23	14	11
	Data	14	16	13	14
	BC-OG	18	25	15	14
	BC	16	13	16	18
Spec	CDG	21	43	10	9
	CDG-FL	18	30	14	10
	SD	14	20	11	11
	SD-FL	14	18	12	11
	Data	51	55	57	42
	BC-OG	53	83	47	30
	BC	41	47	46	31
	CDG	66	141	36	22
	CDG-FL	60	103	45	31
	SD	41	67	31	23
	SD-FL	41	62	36	25

Table 8. Volatility of monthly credit spreads.

5.9 Table 9 and Figures 9–10: Walmart CDS curves

Read:

- Table 9 estimates three models for Walmart: stochastic volatility with jumps (SVJ), stochastic debt (SD), and SVJ plus stochastic debt (SVJ-SD).
- Over 2003–2017, average RMSE is 13.81 bps for SVJ, 9.19 bps for SD, and 2.68 bps for SVJ-SD.
- Excluding 2008, RMSE is 14.24 bps for SVJ, 4.28 bps for SD, and 2.31 bps for SVJ-SD.
- Figure 9 shows that SVJ alone has difficulty matching the average concave CDS curve, while stochastic debt improves the fit.
- Figure 10 shows that jumps and stochastic volatility are useful during 2008, but stochastic debt is important in normal years.

Economic message. The stochastic-debt channel remains useful even when the asset-value process includes stochastic volatility and jumps. Debt dynamics are not merely a substitute for richer asset dynamics.

Table 9

Stochastic volatility-jump model with stochastic debt. For the issuer with most bonds in our sample, Walmart, we estimate a model with stochastic asset volatility and jumps in asset value (SVJ), the stochastic debt model (SD), and a model with stochastic asset volatility, jumps in asset value and stochastic debt (SVJ-SD) by fitting to the year-end CDS premium curves in the years 2003, 2004, ..., 2017. Panel A reports the average parameter estimates where ξ is jump intensity, η^2 is average jump size, γ is jump volatility, λ_A is mean-reversion of volatility, ν_A is mean of volatility, σ_A is volatility of volatility, ρ_V is correlation between volatility and asset value, λ is mean reversion of debt, ν is mean of debt, σ_k is volatility of debt, and ρ_k is correlation between debt and asset value. Panel B reports root-mean-squared errors (RMSEs) overall and across CDS maturities $\frac{1}{2}$, 1, 2, 3, 4, 5, 7, 10, 15, 20 and 30 years.

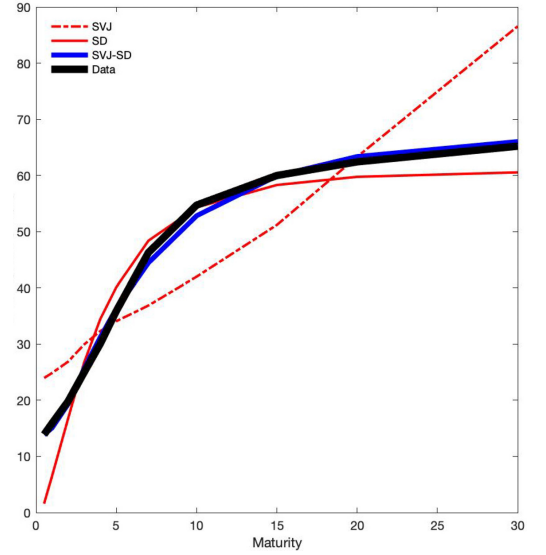
Panel A: Average parameter estimates

	ξ	η^2	γ	λ_A	ν_A	σ_A	ρ_V	λ	ν	σ_k	ρ_k
SVJ	0.010	-0.827	0.064	3.925	0.022	0.062	-0.230	0.314	-1.827	0.451	-0.112
SD											
SVJ-SD	0.010	-0.722	0.043	4.052	0.023	0.315	-0.179	0.249	-1.557	0.230	-0.116

Panel B: Pricing performance (RMSEs)

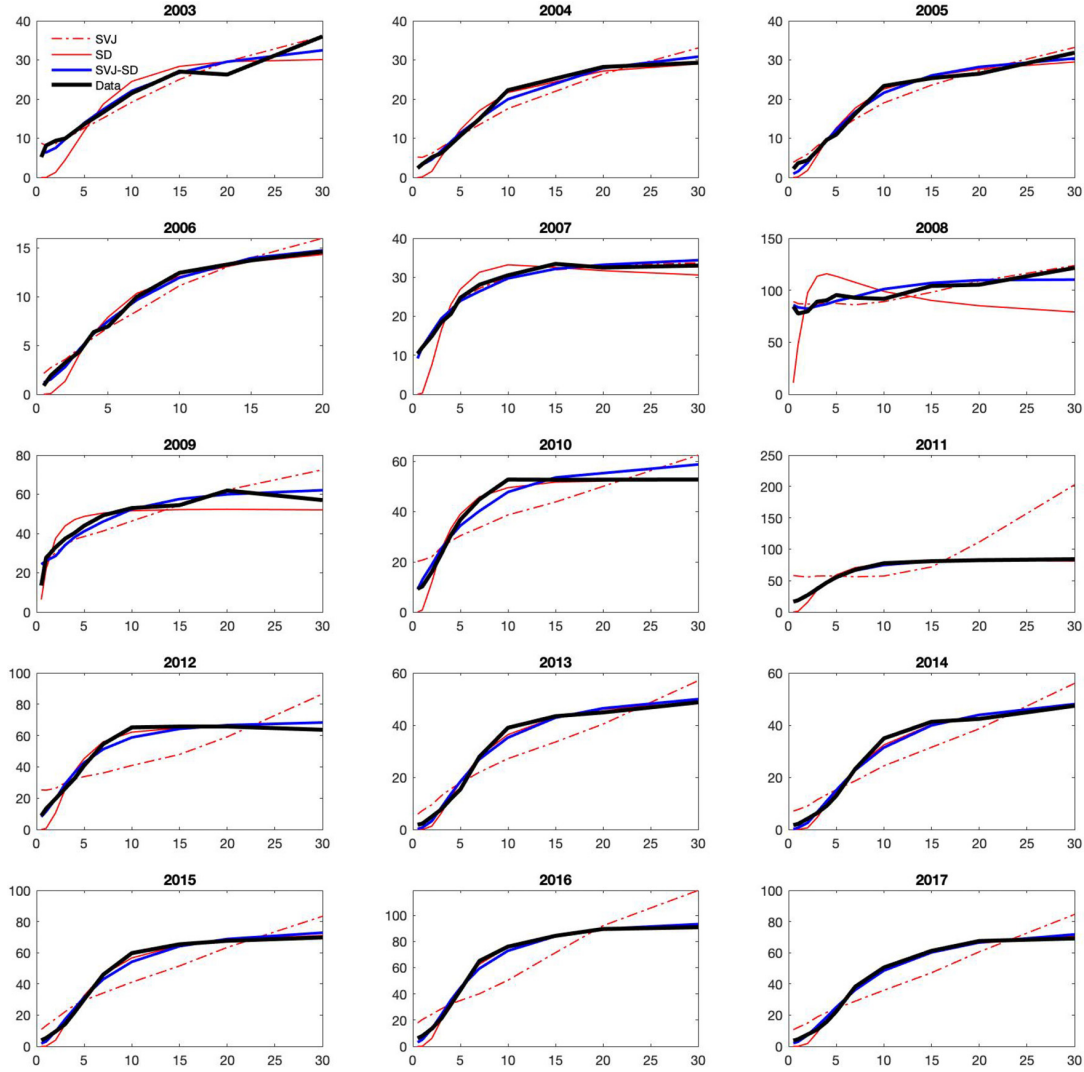
2003–2017												
Maturity	All	0.5	1	2	3	4	5	7	10	15	20	30
SVJ	13.81	13.15	12.01	9.19	6.82	4.21	4.25	10.40	13.60	9.55	8.39	34.05
SD	9.19	20.14	11.06	7.55	6.79	7.60	5.35	4.17	2.75	3.85	6.13	11.60
SVJ-SD	2.68	3.12	2.25	1.72	2.15	2.29	2.30	2.54	4.03	1.42	1.82	4.26
2003–2017 excl. 2008												
SVJ	14.24	13.55	12.17	9.33	7.04	4.29	3.84	10.61	14.06	9.76	8.66	35.33
SD	4.28	7.09	8.20	6.17	2.59	2.83	2.71	1.76	2.11	0.98	2.98	2.55
SVJ-SD	2.31	3.20	1.70	1.64	1.91	2.16	1.83	2.61	3.33	1.24	1.45	3.11

Table 9. SVJ, SD, and SVJ-SD estimates for Walmart.



model with stochastic debt. For the issuer with most bonds in our sample, Walmart, we estimate asset value (SVJ), the stochastic debt model (SD), and a model with stochastic asset volatility, jumps in asset value and stochastic debt (SVJ-SD) by fitting to the year-end CDS premium curves in the years 2003, 2004, ..., 2017. The graph shows the actual CDS curves as well as model-implied yield curves.

Figure 9. Average Walmart CDS curve.



Stochastic volatility-jump model with stochastic debt. For the issuer with most bonds in our sample, Walmart, we estimate a model with stochastic asset volatility and jumps in asset value (SVJ), the stochastic debt model (SD), and a model with stochastic asset volatility, jumps in asset value and stochastic debt (SVJ-SD) by fitting to the year-end CDS premium curves in the years 2003, 2004, ..., 2017. The graph shows the actual CDS curves as well as model-implied yield curves.

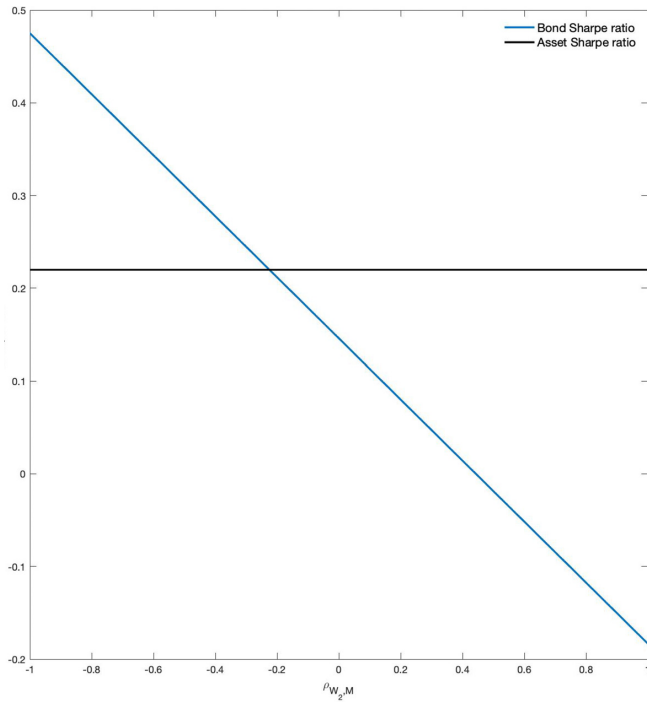
Figure 10. Year-by-year Walmart CDS curves.

5.10 Figure 11, Figure 12, and Table 10: risk premia, jumps in debt, and correlation estimates

Read:

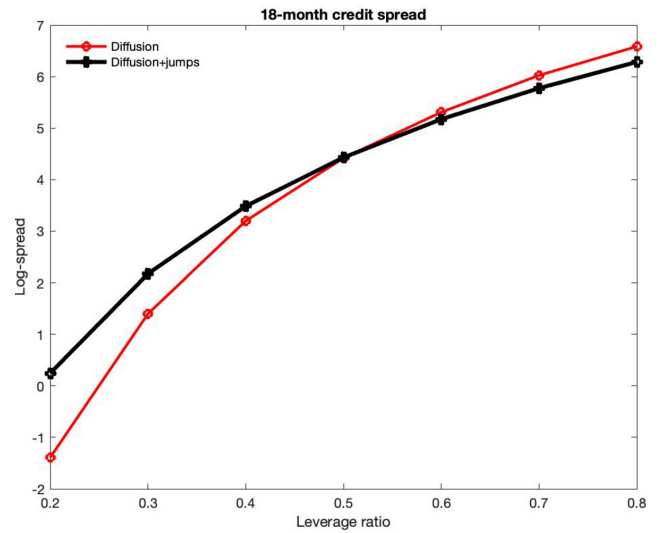
- Figure 11 shows that in the SD model, the bond Sharpe ratio can differ from the asset Sharpe ratio depending on how the debt shock loads on the market.
- To isolate debt-dynamics differences, the paper uses a common bond Sharpe ratio of 0.22 in the main pricing exercise.
- Figure 12 introduces jumps in debt. Jumps raise spreads for low-leverage firms and lower them for high-leverage firms relative to the diffusion SD benchmark.
- Table 10 estimates the debt-asset correlation using discrete debt-equity correlations.
- The data show one-year and two-year correlations of -0.034 and -0.024 , and a three-year correlation of 0.010 .
- The model-implied correlations under the benchmark SD parameters are similar in pattern, and the direct estimate of ρ is -0.1878 , almost identical to the benchmark -0.1868 .

Economic message. The negative debt-asset shock correlation is not just an artifact of the main estimation procedure. It appears again when estimated from debt-equity correlations.



stochastic debt model. The graph plots the asset and bond Sharpe ratios in the stochastic debt model

Figure 11. Sharpe ratios in the stochastic-debt model.



jumps in debt. In the main analysis log-debt in the SD model is given as $dk_t = \lambda(v^Q - l_t)dt + \sigma_k dW_{k,t}$, spreads for this benchmark model ('Diffusion') as well as the SD model where debt can also jump process with jump intensity η and jumps are normally distributed with mean ξ and standard deviation σ_j . The dynamics of assets are given by $dv_t = (r - \delta - \frac{\sigma_v^2}{2})dt + \sigma_v dW_t$ where $r = 0.05$, $\delta = 0.05$, $\sigma = 0.2$, the mean reversion parameter is $\lambda = 0.1814$ while $\sigma_k = 0.2706$ and $v^Q = -0.6593$ in the diffusion case, and $\zeta = 0.0880$ in the jump-diffusion case. The default boundary is 0.7322 in the diffusion case

Figure 12. Credit spreads with jumps in debt.

Table 10

Alternative estimates of debt-asset correlation. The first row (labelled 'Data') shows the 1-, 2-, and 3-year debt-equity correlations using firm-year observations based in firms that have available data in all 30 years of the data sample. The second row ('Benchmark model') shows model-implied correlations between changes in log-level of debt and log-returns of equity in the stochastic debt model, where firm value is given as $dv_t = (\mu - \delta - \frac{\sigma^2}{2})dt + \sigma dW_t$ and log-debt is given as $dk_t = \lambda(v - v_t)dt + \sigma_k dW_{k,t}$ where $\sigma = 0.24$, $\mu = 0.1028$, and $\delta = 0.05$, $\lambda = 0.1814$, $v = -1.0046$, $\sigma_k = 0.2706$ and $\rho = -0.1868$ (where ρ is the correlation between W_t and $W_{k,t}$). The number in the second column is the instantaneous correlation while the next three columns show correlations for 1-, 2-, and 3-year changes. The third row shows the correlation ρ in the benchmark model that minimizes the sum of squared errors between model-implied and data 1-, 2-, and 3-year correlations (where empirical debt-equity correlations proxy for debt-asset correlations). The data is from CRSP/Compustat and the sample period is 1988–2017.

	$T = 0$	$T = 1$	$T = 2$	$T = 3$
Data		−0.034	−0.024	0.010
Benchmark model	−0.1868	−0.104	−0.016	0.075
$\hat{\rho}$ from data	−0.1878			

Table 10. *Alternative estimates of debt-asset correlation.*

6 Short summary

- The paper studies how corporate debt dynamics affect structural credit-risk pricing.
- The key empirical fact is that debt issuance is stochastic and imperfectly correlated with asset-value shocks.
- Low-leverage firms grow debt more; high-leverage firms tend to reduce debt.
- Conditional on equity shocks, low-return firms borrow more in the short run, but high-return firms borrow more in the long run.
- This combination implies short-run pecking-order behavior and long-run target-leverage behavior.
- Standard structural models with constant or deterministic debt miss the short-run stochastic component.
- The SD model adds a stochastic debt shock to mean-reverting debt adjustment.
- This makes leverage volatility exceed asset volatility at short horizons.
- The model is calibrated to debt dynamics and historical default rates, not to observed spreads.
- In out-of-sample spread tests, the SD model performs best, especially for short-maturity investment-grade bonds.
- SD also better matches the volatility of monthly spread changes.
- Adding stochastic debt to a stochastic-volatility/jump model improves the fit to Walmart CDS curves.
- The main lesson is that credit risk depends on debt policy as well as asset value.

7 Conclusion

7.1 Core conclusion

Feldhütter and Schaefer show that corporate debt dynamics are central for credit-risk pricing. Debt is not a passive, fixed boundary. Firms adjust debt over time, and debt adjustment contains stochastic shocks that can move against asset-value shocks in the short run.

7.2 Economic intuition

Default occurs when leverage crosses a boundary. Leverage can rise because firm value falls, because debt rises, or because both happen together. The data show that bad equity-return shocks often coincide with more borrowing in the short run. This raises short-run leverage volatility and changes the pricing of short-term credit risk.

7.3 Takeaway

The paper moves structural credit-risk models closer to observed corporate financing behavior. Its main message is that modeling asset-value dynamics alone is not enough: a realistic default boundary must include stochastic debt dynamics.